[Your report title, e.g. Predicting FIFA World Cup 2026 Match Outcomes with Linear Regression]

HIT140 Foundations of Data Science · Assessment 3 · Sydney Group 07 · Code: [GitHub/OneDrive URL]

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1. Introduction and Data

[Write in your own words: aim of Objective 2 and the two regression tasks; data sources (see data/raw/SOURCES.md); how 104/208 rows were compiled; the 'known before kick-off' rule and how it was enforced; score convention (extra time counted, shoot-out = draw); validation against the 12 group tables.]

Table 1. Explanatory variables (all known before kick-off)

Concept	LR 2.1 variable (match level)	LR 2.2 variable (team level)	Source
FIFA ranking	fifa_rank_diff (team1 − team2)	fifa_rank, opp_fifa_rank	FIFA ranking, 11 Jun 2026
Squad market value	log_value_ratio = ln(V1/V2)	log_market_value, opp_log_market_value	Transfermarkt
Average squad age	avg_age_diff	—	RotoWire
Host nation	host_diff (+1/0/−1)	is_host	CAN, MEX, USA
World Cup experience	wc_experience_diff	—	Wikipedia
Confederation strength	elite_conf_diff (UEFA/CONMEBOL)	—	FIFA
Rest days	rest_days_diff	—	Match schedule
Tournament form (earlier matches only)	form_gd_pg_diff	form_gf_pg, opp_form_ga_pg	Match results
Stage	—	is_knockout	Match schedule
Target	goal_difference (104 rows)	goals (208 rows)	

2. Linear Regression 2.1 — Goal Difference

2.1 Exploration and model building

[Write in your own words: target distribution (mean 0.48, SD 2.02); strongest correlations (value ratio r = 0.66, rank r = −0.62); collinearity between value and rank (r = −0.79, VIF 7.4); Model A with all 8 variables; backward elimination to Model B.]

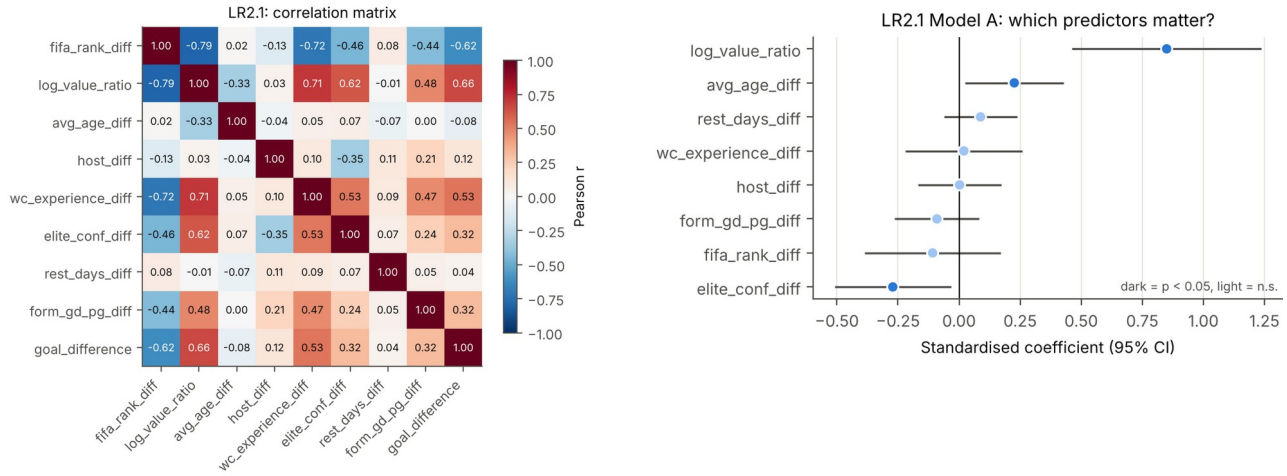


Figure 1. LR 2.1 — correlation matrix (left) and standardised coefficients of Model A with 95% CIs (right)

2.2 Evaluation and interpretation

[Write in your own words: compare Model A vs B using Table 2 (overfitting of A; B better on unseen data; both beat the mean baseline); interpret the Model B coefficients in Table 3 in plain football terms; discuss the negative elite_conf_diff carefully (no causal claim); residual checks (assumptions reasonably met).]

Table 2. Model comparison (A = all 8 variables; B = backward elimination, p < 0.05; C = A with log(1 + goals) target, predictions back-transformed)

Metric	2.1 Model A	2.1 Model B	2.2 Model A	2.2 Model B	2.2 Model C (log)
Explanatory variables used	8	3	8	2	8
R^2 / adj. R^2 (all rows)	0.511 / 0.469	0.495 / 0.480	0.295 / 0.267	0.268 / 0.261	0.272 / 0.242 (log)
F-test p-value	<0.001	<0.001	<0.001	<0.001	<0.001

Metric	2.1 Model A	2.1 Model B	2.2 Model A	2.2 Model B	2.2 Model C (log)
Test R^2 (80/20 split)	0.284	0.495	0.149	0.180	0.247
Test RMSE / MAE	1.78 / 1.44	1.50 / 1.16	1.02 / 0.81	1.01 / 0.78	0.96 / 0.73
Test NRMSE	0.223	0.187	0.205	0.201	0.193
Baseline (mean) test RMSE	2.11	2.11	1.18	1.18	1.18
10-fold CV (×20) R^2	0.261	0.336	0.104	0.142	0.093
10-fold CV RMSE (baseline)	1.53 (1.98)	1.45 (1.98)	1.21 (1.37)	1.19 (1.37)	1.24 (1.37)
Shapiro–Wilk p / Breusch–Pagan p	0.60 / 0.15		0.001 / <0.001		0.021 / –

Table 3. Final (Model B) coefficients

Model	Variable	Coef.	95% CI	p-value	Std. coef.
2.1-B	log_value_ratio	1.089	[0.841, 1.336]	<0.001	0.90
2.1-B	avg_age_diff	0.321	[0.101, 0.540]	0.005	0.24
2.1-B	elite_conf_diff	−0.706	[−1.231, −0.180]	0.009	−0.26
2.2-B	log_market_value	0.448	[0.312, 0.585]	<0.001	0.39
2.2-B	opp_fifa_rank	0.020	[0.013, 0.027]	<0.001	0.33

3. Linear Regression 2.2 — Goals Scored by a Team

3.1 Exploration and model building

[Write in your own words: target is a right-skewed count (mean 1.48, 0–7 goals); why both own and opponent variables are needed; weaker correlations than 2.1; VIF all < 3.2; Model A and backward elimination to Model B.]



Figure 2. LR 2.2 — distribution of goals per team per match (left) and standardised coefficients of Model A (right)

3.2 Evaluation and interpretation

[Write in your own words: compare Model A vs B (Table 2); interpret Model B (Table 3), e.g. what a 50-place weaker opponent means; diagnostics show non-normal, heteroscedastic residuals because goals are counts (Figure 3); Model C (log(1 + goals) target) as an improvement attempt: better residual normality and test-split error, but no gain in cross-validation; Poisson regression as future work.]

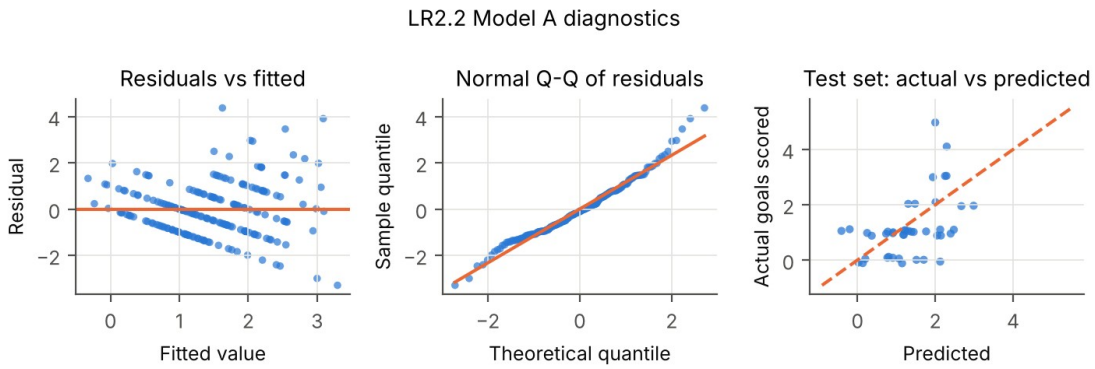


Figure 3. LR 2.2 Model A — residuals vs fitted, normal Q-Q plot, and actual vs predicted on the test set

4. Discussion and Conclusion

[Write in your own words: why 2.1 explains more variance than 2.2; squad value as the most consistent predictor; limitations (small n, Transfermarkt snapshot date, form = 0 in first match, arbitrary team1/team2 order, paired rows in 2.2, ordinal ranking, secondary sources for knockout scores); improvements and conclusion.]

References

[Write in your own words: list the sources from data/raw/SOURCES.md in the referencing style your unit requires.]